

AttendAI — Offline Smart Attendance Platform

Full System Operating Guide & 2-Factor AI Architecture Documentation

1. System Overview & Core Concept

AttendAI is a locally hosted, AI-assisted classroom attendance platform built for offline environments. Students present an ID card and face the camera; local AI reads the card, verifies student identity, performs live face embedding verification against local SQLite records, enforces teacher session time windows, and updates the real-time dashboard instantly via WebSockets — with **100% zero cloud dependency**.

2. Two-Factor Identity Verification Pipeline

To prevent fraudulent attendance (such as one student holding another person's card), AttendAI enforces a strict 2-Factor verification flow:

- **Factor 1 (ID Card OCR):** OpenCV contour detection + perspective warp + EasyOCR extracts Student ID.
- **Factor 2 (Face Vector Verification):** Generates a 128-dimensional normalized face embedding and computes cosine similarity against the enrolled reference embedding (threshold $\geq 65\%$).

Verdict Output: ID Card: Verified ✓ | Face: Matched ✓ | Confidence: 96.8% | PRESENT

3. Demo Credentials & Access URLs

Resource	URL / Value
Web Application UI	http://localhost:5173
Backend API Endpoint	http://127.0.0.1:8000
Teacher Login Email	teacher@apex.edu
Teacher Password	admin123 (Or click 1-Click Quick Demo Login)

4. Step-by-Step Teacher Operating Guide

1. **Launch App:** Run START_SMART_ATTENDANCE.bat or open http://localhost:5173.
2. **Login:** Log in as teacher or click *Quick Demo Login*.
3. **Classes & Enrollment:** Enroll students, upload face photos to generate local embeddings, and print student ID cards.
4. **Start Session:** Create an attendance session for a class (e.g. CSE-A, Data Structures) and set status to ACTIVE.
5. **Live Scanner Studio:** Open camera feed or click *Simulate Instant 2-Factor AI Verification Scan*.
6. **Auto-Absence:** Closing a session automatically marks all un-scanned students as ABSENT.
7. **Teacher Override:** Perform audit-logged status overrides with mandatory reason recording.
8. **Export Reports:** One-click export daily/monthly reports in PDF, Excel (.xlsx), and CSV formats.

5. Direct System Download Endpoints

Report Type	Direct Download Link
Live Attendance PDF Report	http://127.0.0.1:8000/api/reports/export/pdf
Excel (.xlsx) Attendance Report	http://127.0.0.1:8000/api/reports/export/excel

CSV Attendance Report	http://127.0.0.1:8000/api/reports/export/csv
System Documentation PDF	http://127.0.0.1:8000/uploads/AttendAI_System_Guide.pdf